

HOW DOES THE SATURATION OF LIPIDS INFLUENCE THE ALIGNMENT OF MEMBRANE PEPTIDES?

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Two well-known antimicrobial peptides (AMPs) are magainin and PGLa enhancing their bioactivity when combined is referred to as synergism, which is most effective at a 1:1 molar ratio.¹ Further studies show that the alignment of the peptides on the surface of cell membranes depends on the saturation of the lipid fatty acid chains within the membrane.^{1,2} Saturated lipids pushes the equilibrium towards a transmembrane orientation whereas unsaturated lipids stabilize an in-planar orientation of PGLa and the model peptide LAH4.^{1,2} We are investigating the effect of the peptides on the properties of saturated and non-saturated lipids. The perturbation capability of the peptides on the membrane can be characterized by the order parameter.³

Generally, a high order parameter corresponds to a well-ordered structure, while a low value indicates reduced order. The membrane models used in this study were composed of both pure and mixed lipids (e.g., POPE, POPG, DMPE, and DMPG). Previously exclusively the saturated aryl chain(s) was measured by deuterium NMR. In-planar peptides show a more pronounced effect on the order parameter than transmembrane peptides.²

For non-saturated lipids the energetic penalty from chain dis-ordering for hosting in-planar peptides is probably lower than for saturated lipids. In order to test this hypothesis, we need access to the order parameter of the non-saturated chain. This can be achieved by measuring time averaged ¹³C-¹H dipolar coupling using the natural abundance of ¹³C. The R-type Proton-Detected Local Field (R-PDLF) pulse sequence⁴ enables atomistic resolution in the isotropic chemical shift dimension allowing to assign the dipolar coupling in the second dimension.^{1,2}

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